

Tejas Goyal

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EDUCATION

Carnegie Mellon University

Master of Information Systems

Pittsburgh, PA

August 2025 – December 2026

- Courses: LLM Methods and Applications, LLM Systems, AI Agents, Deep Learning

PROFESSIONAL EXPERIENCE

Qualcomm

Machine Learning Engineer Intern

San Diego, CA

May 2026 – Present

- Built a hybrid retrieval engine over a Neo4j knowledge graph of 5G/LTE diagnostic expertise, fusing BM25, bi-encoder cosine similarity, and cross-encoder reranking via Reciprocal Rank Fusion to power chat, workflow, and agent tools across **10K+ modem logs processed daily** at petabyte scale
- Layered an automated diagnosis agent that pulls code recipes from ranked graph nodes, runs them in sandboxed auto-scaling EC2, and returns a root-cause diagnosis; flattened branching logic and pre-computed conditions so a single on-prem **31B dense model (Gemma 4)** cleared the full daily load within token budget

OpenHands

Machine Learning Engineer

Pittsburgh, PA

August 2025 – Present

- Engineered a configurable timeout system for MCP servers preventing tool execution failures during long-running agent operations, improving reliability for **65,000+ active developers** [PR]
- Translated enterprise client requirements into open-source agent SDK features, benchmarking the roadmap against Cursor, Jules, Devin, and Codex to drive the project's technical direction

Confluent

Technical Program Manager

Bangalore, India

January 2024 – August 2025

- Drove the roadmap for an AI-powered Customer Intelligence Hub, aligning engineering and go-to-market teams to ship a **40% faster** reporting pipeline for **500+ sales representatives**

Philips

AI Engineer Intern

Bangalore, India

June 2023 – August 2023

- Built an LLM patient-support assistant on Azure OpenAI with a context-aware RAG pipeline (Pinecone, spaCy chunking), cutting query resolution time **60%** for a projected **\$1.1M** in savings

RESEARCH EXPERIENCE

TEEL Lab | CMU (Advisor: Prof. Jaromir Savelka)

December 2025 – Present

- Proposed **spec learning**, a four-stage compiler that distills **20 preference pairs** into a natural-language system prompt for inference-time LLM alignment with no weight updates; outperforms LoRA-DPO trained on **1,000 pairs** across **6 domains** with a **75% pairwise win rate** (*NeurIPS 2026, under review*)
- Benchmarked **8 LLMs** on labeling domain-specific text spans across **5 datasets**, showing that grounding prompts in a structured definition of each label lifts exact-match F1 from **1.8% to 57.5%** on expert tasks, with a small **3–5%** drop on simple pattern matching (*ACL 2026 SRW, San Diego*)

SELECT PUBLICATIONS

- Towards Spec Learning: Inference-Time Alignment from Preference Pairs.** *NeurIPS 2026* [paper]
- Semantic Span Annotation: An Exploratory Study of LLM Annotation.** *ACL 2026 SRW* [paper]

PROJECTS

SGLang – WLFU Radix Cache Eviction Policy | Open-Source Contribution (28K stars)

- Designed WLFU, a radix cache eviction policy extending LRU with a log-frequency decay factor that de-prioritizes high-hit prefixes to increase cache availability for underserved shared prefixes in finite workloads
- Achieved **22.8% reduction in mean TTFT** and **12.8% in mean E2E latency** vs. LRU on Qwen2.5-7B (A100); contributed to SGLang (deployed at Cursor, NVIDIA, and xAI)

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, TypeScript

ML: PyTorch, HuggingFace Transformers, scikit-learn, NumPy, vLLM, SGLang

Tools & Cloud: AWS, Azure, Docker, Git, Neo4j, PostgreSQL